



June 17, 2026

PhD Admissions, University of Southampton

Dear PhD Admissions Committee,

I am writing to express my strong interest in pursuing a PhD in the development of a novel tunable SiC entangled photon source at the University of Southampton. As an AI Data Science Engineer with a passion for quantum computing and nano-opto-electro-mechanical systems, I believe that this project aligns perfectly with my research interests and goals.

My experience working on the "Cardiovascular Medical Digital Twin" project has equipped me with expertise in developing novel algorithms and models that can be applied to the field of quantum entanglement. Specifically, my work on the linear-time state space model, *MedicalMambaTS*, *has shown promising results in achieving sub-millisecond bedside inference for cardiovascular disease diagnosis. This achievement demonstrates my ability to develop fast and*

Furthermore, my experience with 0D Windkessel Physics-Informed Operator Network using PyTorch Autograd, *PIDeepONet*, *has given me a deep understanding of physical constraints that can be applied to quantum systems. This knowledge*

I am confident that my skills and experience make me an ideal candidate for this PhD project. I am excited about the opportunity to work with the University of Southampton's team and contribute to the development of a novel tunable SiC entangled photon source.

Thank you for considering my application.

Sincerely,

Sincerely,

Ismail Aissaoui